

ECON2050 Tutorial 5

Linear Algebra I

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Today's Plan

- ▶ Tutorial questions: Q1–Q4

Question 1

For each set, determine (a) whether it is linearly independent or linearly dependent, and (b) its span.

(i)

$$S_1 = \{\mathbf{v}_1, \mathbf{v}_2\}, \quad \mathbf{v}_1 = (1, -2), \quad \mathbf{v}_2 = (2, -4).$$

(ii)

$$S_2 = \{\mathbf{v}_1, \mathbf{v}_2\}, \quad \mathbf{v}_1 = (1, 0), \quad \mathbf{v}_2 = (0, 1).$$

(iii)

$$S_3 = \{\mathbf{v}_1, \mathbf{v}_2\}, \quad \mathbf{v}_1 = (0, 1, 0), \quad \mathbf{v}_2 = (-2, 2, 0).$$

(iv)

$$S_4 = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}, \quad \mathbf{v}_1 = (0, 1, 0), \quad \mathbf{v}_2 = (-2, 2, 0), \quad \mathbf{v}_3 = (-2, 3, 0).$$

(v)

$$S_5 = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}, \quad \mathbf{v}_1 = (1, 1, 0), \quad \mathbf{v}_2 = (0, -3, 1), \quad \mathbf{v}_3 = (2, 4, -1).$$

Recap: Linear combinations and span

Linear combination

Let $S = \{\mathbf{x}^1, \dots, \mathbf{x}^n\} \subseteq \mathbb{R}^m$. A **vector** $\mathbf{y}$ is a linear combination of vectors in S if there exist scalars $a_1, \dots, a_n$ such that

$$\mathbf{y} = \sum_{j=1}^n a_j \mathbf{x}^j.$$

Span

The set of all linear combinations of vectors in S is the **span** of S :

$$\text{span}(S) = \left\{ \sum_{j=1}^n a_j \mathbf{x}^j : a_1, \dots, a_n \in \mathbb{R} \right\}.$$

Recap: Linear independence and dependence

Linear independence

A set $S = \{\mathbf{x}^1, \dots, \mathbf{x}^n\}$ is **linearly independent** if

$$\sum_{j=1}^n a_j \mathbf{x}^j = \mathbf{0} \implies a_j = 0 \text{ for all } j.$$

Linear dependence

The set is **linearly dependent** if there exist scalars, not all zero, such that

$$\sum_{j=1}^n a_j \mathbf{x}^j = \mathbf{0}.$$

Equivalently, at least one vector can be expressed as a linear combination of the others.

Tutorial Q1: Knowledge used

Knowledge used

- ▶ **Lecture 5:** linear combinations and span.
- ▶ **Lecture 5:** linear independence and dependence.
- ▶ To test independence, solve

$$a_1 \mathbf{v}_1 + \cdots + a_k \mathbf{v}_k = \mathbf{0}.$$

The set is linearly independent exactly when the only solution is $a_1 = \cdots = a_k = 0$.

- ▶ The span is the set of all linear combinations of the vectors in the set.

Tutorial Q1(i): Solution

Solution

Since

$$\mathbf{v}_2 = (2, -4) = 2(1, -2) = 2\mathbf{v}_1,$$

S_1 is linearly dependent.

Its span is

$$\begin{aligned}\text{span}(S_1) &= \{a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 : a_1, a_2 \in \mathbb{R}\} \\ &= \{(a_1 + 2a_2)(1, -2) : a_1, a_2 \in \mathbb{R}\} \\ &= \{k(1, -2) : k \in \mathbb{R}\}.\end{aligned}$$

Therefore,

$$\text{span}(S_1) = \{\mathbf{z} = (x, y) \in \mathbb{R}^2 : y = -2x\}.$$

Tutorial Q1(ii): Solution

Solution

Consider

$$a_1(1, 0) + a_2(0, 1) = (0, 0).$$

This is equivalent to

$$a_1 = 0, \quad a_2 = 0.$$

Hence S_2 is linearly independent.

Moreover,

$$\begin{aligned}\text{span}(S_2) &= \{a_1(1, 0) + a_2(0, 1) : a_1, a_2 \in \mathbb{R}\} \\ &= \{(a_1, a_2) : a_1, a_2 \in \mathbb{R}\} = \mathbb{R}^2.\end{aligned}$$

Tutorial Q1(iii): Solution

Solution

Consider

$$a_1(0, 1, 0) + a_2(-2, 2, 0) = (0, 0, 0).$$

The left-hand side is

$$(-2a_2, a_1 + 2a_2, 0).$$

Thus

$$-2a_2 = 0, \quad a_1 + 2a_2 = 0,$$

which implies $a_1 = a_2 = 0$. Therefore, S_3 is **linearly independent**.

Tutorial Q1(iii): Solution — span

Solution

The span is

$$\text{span}(S_3) = \{(-2a_2, a_1 + 2a_2, 0) : a_1, a_2 \in \mathbb{R}\}.$$

The third component is always zero. Conversely, any **vector**

$$\mathbf{z} = (x, y, 0)$$

can be generated by choosing

$$a_2 = -\frac{x}{2}, \quad a_1 = y + x.$$

Hence

$$\text{span}(S_3) = \{\mathbf{z} = (x, y, 0) : x, y \in \mathbb{R}\},$$

the xy -plane.

Tutorial Q1(iv): Solution

Solution

Here

$$(-2, 3, 0) = (0, 1, 0) + (-2, 2, 0),$$

or

$$\mathbf{v}_3 - \mathbf{v}_1 - \mathbf{v}_2 = \mathbf{0}.$$

Therefore, S_4 is linearly dependent.

Also,

$$S_4 = S_3 \cup \{\mathbf{v}_3\}, \quad \mathbf{v}_3 \in \text{span}(S_3).$$

Adding $\mathbf{v}_3$ therefore does not enlarge the span. Thus

$$\text{span}(S_4) = \text{span}(S_3) = \{\mathbf{z} = (x, y, 0) : x, y \in \mathbb{R}\}.$$

Tutorial Q1(v): Solution — independence setup

Solution

To test linear independence, consider

$$a_1(1, 1, 0) + a_2(0, -3, 1) + a_3(2, 4, -1) = (0, 0, 0).$$

This is equivalent to

$$a_1 + 2a_3 = 0,$$

$$a_1 - 3a_2 + 4a_3 = 0,$$

$$a_2 - a_3 = 0.$$

Tutorial Q1(v): Solution — independence conclusion

Solution

From

$$a_2 - a_3 = 0,$$

we have $a_2 = a_3$. Substituting this into

$$a_1 - 3a_2 + 4a_3 = 0$$

gives

$$a_1 + a_3 = 0.$$

Comparing this with $a_1 + 2a_3 = 0$ gives $a_3 = 0$, and hence

$$a_1 = a_2 = a_3 = 0.$$

Therefore, S_5 is linearly independent.

Tutorial Q1(v): Solution — span

Solution

A general linear combination is

$$a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + a_3 \mathbf{v}_3 = (a_1 + 2a_3, a_1 - 3a_2 + 4a_3, a_2 - a_3).$$

For an arbitrary **vector** $\mathbf{x} = (x_1, x_2, x_3) \in \mathbb{R}^3$, solve

$$\begin{aligned} a_1 + 2a_3 &= x_1, \\ a_1 - 3a_2 + 4a_3 &= x_2, \\ a_2 - a_3 &= x_3. \end{aligned}$$

One solution is

$$\begin{aligned} a_1 &= -x_1 + 2x_2 + 6x_3, \\ a_2 &= x_1 - x_2 - 2x_3, \quad a_3 = x_1 - x_2 - 3x_3. \end{aligned}$$

Thus every $\mathbf{x} \in \mathbb{R}^3$ is generated, so

$$\text{span}(S_5) = \mathbb{R}^3.$$

Question 2

Determine the rank of the following matrices:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 2 \\ 1 & -3 & 4 \\ 0 & 1 & -1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 1 & 0 \\ -2 & 2 & 0 \\ -2 & 3 & 0 \end{bmatrix},$$

$$\mathbf{C} = \begin{bmatrix} 2 & 0 & -1 & 3 \\ 1 & 1 & 2 & 2 \\ 2 & 0 & -1 & 1 \end{bmatrix}.$$

Recap: Rank

Rank of a set of vectors

The **rank** of a set S of vectors is the maximal number of linearly independent vectors in S :

$$\text{rank}(S).$$

If $S \subseteq \mathbb{R}^m$, then $\text{rank}(S) \leq m$.

Rank of a matrix

The column rank of $\mathbf{A}$ is the maximal number of linearly independent columns; the row rank is defined analogously. For every matrix,

$$\text{column rank} = \text{row rank}.$$

This common value is the **rank of the matrix**, denoted $\text{rank}(\mathbf{A})$.

Tutorial Q2: Knowledge used

Knowledge used

- ▶ **Lecture 5:** the rank of a set of vectors is the maximal number of linearly independent vectors.
- ▶ **Lecture 5:** for a matrix, column rank equals row rank.
- ▶ Hence the rank of a matrix can be found using either its columns or its rows.
- ▶ Results from Q1 can be reused when the rows or columns coincide with the vector sets studied there.

Tutorial Q2: Solution — $\mathbf{A}$ and $\mathbf{B}$

Solution

The columns of $\mathbf{A}$ are exactly the three vectors in S_5 from Q1(v). Since S_5 is linearly independent,

$$\text{rank}(\mathbf{A}) = 3.$$

The rows of $\mathbf{B}$ are exactly the three vectors in S_4 from Q1(iv). We showed that the maximal number of linearly independent vectors in S_4 is two. Since row rank equals column rank,

$$\text{rank}(\mathbf{B}) = 2.$$

Tutorial Q2: Solution — $\mathbf{C}$ setup

Solution

Use the rows of $\mathbf{C}$:

$$\mathbf{c}_1 = (2, 0, -1, 3), \quad \mathbf{c}_2 = (1, 1, 2, 2), \quad \mathbf{c}_3 = (2, 0, -1, 1).$$

Consider

$$a_1 \mathbf{c}_1 + a_2 \mathbf{c}_2 + a_3 \mathbf{c}_3 = \mathbf{0}.$$

Component by component, this gives

$$2a_1 + a_2 + 2a_3 = 0,$$

$$a_2 = 0,$$

$$-a_1 + 2a_2 - a_3 = 0,$$

$$3a_1 + 2a_2 + a_3 = 0.$$

Tutorial Q2: Solution — $\mathbf{C}$ conclusion

Solution

From $a_2 = 0$, the first equation gives

$$a_1 = -a_3.$$

The fourth equation gives

$$3a_1 = -a_3.$$

Combining these relations,

$$-3a_3 = -a_3 \implies a_3 = 0.$$

Hence $a_1 = 0$ and $a_2 = 0$. Therefore the three rows are linearly independent, so the row rank is three. Thus

$$\text{rank}(\mathbf{C}) = 3.$$

Question 3

Let $\mathbf{A}$ be an $m \times n$ matrix and let $\mathbf{b} \in \mathbb{R}^m$.

Carefully argue why

“there exists $\mathbf{x} \in \mathbb{R}^n$ such that $\mathbf{Ax} = \mathbf{b}$ ”

is equivalent to

“ $\mathbf{b} \in \text{span}(\mathbf{A})$ ”.

Also give an example with $m = 2$ and $n = 3$.

Recap: Matrix–vector multiplication and span

$\mathbf{Ax}$

Let $\mathbf{A}$ be an $m \times n$ matrix and let $\mathbf{x} \in \mathbb{R}^n$ be an $n \times 1$ vector. Then $\mathbf{Ax} \in \mathbb{R}^m$, and its i -th component is

$$(\mathbf{Ax})_i = \sum_{j=1}^n a_{ij}x_j = \mathbf{a}_i \cdot \mathbf{x},$$

where $\mathbf{a}_i$ is the i -th row of $\mathbf{A}$.

Span

The set of all linear combinations of vectors in S is the **span** of S :

$$\text{span}(S) = \left\{ \sum_{j=1}^n a_j \mathbf{x}^j : a_1, \dots, a_n \in \mathbb{R} \right\}.$$

$\text{span}(\mathbf{A})$ denotes the span of the columns of $\mathbf{A}$.

Tutorial Q3: Knowledge used

Knowledge used

- ▶ **Lecture 5:** matrix–vector multiplication.
- ▶ If the columns of $\mathbf{A}$ are the **vectors** $\mathbf{a}^1, \dots, \mathbf{a}^n$, then

$$\mathbf{Ax} = x_1 \mathbf{a}^1 + \dots + x_n \mathbf{a}^n.$$

- ▶ **Lecture 5:** $\text{span}(\mathbf{A})$ means the span of the columns of $\mathbf{A}$.

Tutorial Q3: Solution — matrix as columns

Solution

Write the matrix by columns:

$$\mathbf{A} = [\mathbf{a}^1 \quad \mathbf{a}^2 \quad \cdots \quad \mathbf{a}^n], \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

By the definition of matrix–vector multiplication,

$$\mathbf{Ax} = x_1 \mathbf{a}^1 + x_2 \mathbf{a}^2 + \cdots + x_n \mathbf{a}^n.$$

Tutorial Q3: Solution — $\mathbf{Ax} = \mathbf{b}$ implies span

Solution

The equation

$$\mathbf{Ax} = \mathbf{b}$$

can therefore be written as

$$x_1 \mathbf{a}^1 + x_2 \mathbf{a}^2 + \cdots + x_n \mathbf{a}^n = \mathbf{b}.$$

This means precisely that $\mathbf{b}$ can be written as a linear combination of the columns of $\mathbf{A}$. Hence

$$\mathbf{Ax} = \mathbf{b} \implies \mathbf{b} \in \text{span}(\mathbf{A}).$$

Tutorial Q3: Solution — converse

Solution

Conversely, suppose

$$\mathbf{b} \in \text{span}(\mathbf{A}).$$

By definition, there exist scalars $x_1, \dots, x_n$ such that

$$\mathbf{b} = x_1 \mathbf{a}^1 + x_2 \mathbf{a}^2 + \dots + x_n \mathbf{a}^n.$$

Define the **vector**

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

Then the right-hand side is exactly $\mathbf{Ax}$. Therefore,

$$\mathbf{b} \in \text{span}(\mathbf{A}) \implies \exists \mathbf{x} \in \mathbb{R}^n : \mathbf{Ax} = \mathbf{b}.$$

Thus the two statements are equivalent.

Tutorial Q3: Solution — example

Solution

Take

$$\mathbf{A} = \begin{bmatrix} 4 & 1 & -2 \\ 1 & 0 & 5 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}.$$

Then

$$\mathbf{Ax} = \mathbf{b}$$

is equivalent to

$$x_1 \begin{bmatrix} 4 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -2 \\ 5 \end{bmatrix} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}.$$

This is exactly the statement that the **vector** $\mathbf{b}$ is in the span of the three columns of the **matrix** $\mathbf{A}$.

Question 4

Consider the system

$$\mathbf{Ax} = \mathbf{b},$$

where

$$\mathbf{A} = \begin{bmatrix} 4 & 1 & -2 \\ 1 & 0 & 5 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}.$$

Determine whether the system has a solution.

Recap: Solvability of linear equations

Alternative theorem

Let $\mathbf{A}$ be an $m \times n$ matrix, $\mathbf{b} \in \mathbb{R}^m$, and

$$F = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{Ax} = \mathbf{b}\}.$$

Then exactly one of the following holds:

1. $F \neq \emptyset$; or
2. there exists a **vector** $\mathbf{y} \in \mathbb{R}^m$ such that

$$\mathbf{yA} = \mathbf{0}, \quad \mathbf{y} \cdot \mathbf{b} \neq 0.$$

Interpretation

The second alternative provides a set of weights on the equations that derives an inconsistency, proving that the system has no solution.

Tutorial Q4: Knowledge used

Knowledge used

► **Lecture 5:** solvability of linear equations.

► Exactly one of the following can hold:

1. there exists $\mathbf{x}$ such that $\mathbf{Ax} = \mathbf{b}$; or
2. there exists $\mathbf{y}$ such that

$$\mathbf{yA} = \mathbf{0}, \quad \mathbf{y} \cdot \mathbf{b} \neq 0.$$

► Therefore, if every $\mathbf{y}$ satisfying $\mathbf{yA} = \mathbf{0}$ also satisfies $\mathbf{y} \cdot \mathbf{b} = 0$, the system must be solvable.

Tutorial Q4: Solution

Solution

Let $\mathbf{y} = (y_1, y_2)$ and impose

$$\mathbf{yA} = \mathbf{0}.$$

Then

$$(y_1, y_2) \begin{bmatrix} 4 & 1 & -2 \\ 1 & 0 & 5 \end{bmatrix} = (4y_1 + y_2, y_1, -2y_1 + 5y_2) = (0, 0, 0).$$

The second component gives $y_1 = 0$, and then the first gives $y_2 = 0$. Thus the only possible **vector** is

$$\mathbf{y} = \mathbf{0},$$

for which

$$\mathbf{y} \cdot \mathbf{b} = 0.$$

Hence no inconsistency certificate exists. By the solvability theorem,

the system has a solution.